Matching & weighting

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1 Recap: conditional independence / covariate adjustment

- Can treatment allocation be randomized? – No
- Time series available? – No
- Covariates available? – Yes
- Regression adjustment
- Matching and weighting
- Double ML
- Core assumptions: conditional independence / unconfoundedness

1.1 Non-random assignment – quasi-experimental methods

- We have non-random assignment → there are selection effects:
  - **Selection on unobservables**
    - We do not have sufficient covariates to capture the selection mechanism
    - Focal treatment effect correlates with the unobserved error because of these omitted variables / confounders (i.e., endogeneity bias, confounder bias)
    - Instrumental variable methods
    - IV-free methods
  - **Selection on observables**
    - We have ALL necessary covariates to capture the selection mechanism
    - **Conditional independence** – when we sufficiently control for the influence of the covariates, we achieve independence / unconfoundedness of the treatment variable
    - Regression adjustment, matching and weighting
    - Double ML

1.2 Recap: non-random assignment

- We do not have random assignment (i.e., independence of potential outcomes from treatment $\{Y_i(1), Y_i(0) \perp T_i\}$) but we assume conditional independence: $\{Y_i(1), Y_i(0)\} \perp T_i | X_i$
- In lec 2, we have learned that OLS regression can identify ATE by controlling for X
  - However, there are some problems with OLS: it is linear, parametric, and does a variance weighting
- We want to look at other ways of estimating treatment effects (i.e., ATE, ATET) relying on the principle of estimating treatment effects for individuals that are equal on X

Data rich environments: many of the following methods are especially popular in data-rich environments, like online advertising. We have a massive feature / covariate space. They typically assume that we have a high-dimensional covariate matrix (many covariates). Traditional linear regression is good when focusing on a few theoretically well justified vairables and mainly linear influences (or specific non-linear or interaction terms): it struggles to control for hundreds of complex, interacting variables without overfitting.

2 Core idea of matching

Situation

- A (set of) confounding variable(s) X assign individuals to treatment and control so that these two groups are initially not comparable in their potential outcomes

Idea

- We want to pair treated units with an equal/similar untreated unit to achieve a new sample that is comparable
- We want to find statistical twins
- We then estimate differences (treatment effects) only for matched individuals comparable on X

How do we find equal/similar treated and untreated units?

2.1 A simplified (ideal) example

Treatment group

unit	age	spending
1	28	17700
2	34	10200
3	29	14400
4	25	20800
5	29	6100
6	23	28600
7	33	21900
8	27	28800
9	31	20300
10	26	28100
11	25	9400
12	27	14300
13	29	12500
14	24	19700
15	25	10100
16	43	10700
17	28	11500
18	27	10700
19	28	16300
Average:	28.5	16426

Control group

unit	age	spending
1	43	20900
2	50	31000
3	30	21000
4	27	9300
5	54	41100
6	48	29800
7	39	42000
8	28	8800
9	24	25500
10	33	15500
11	26	400
12	31	26600
13	26	16500
14	34	24200
15	25	23300
16	24	9700
17	29	6200
18	35	30200
19	32	17800
20	23	9500
21	32	25900
Average:	33	20724

Matched control group

unit	age	spending
8	28	8800
14	34	24200
17	29	6200
15	25	23300
17	29	6200
20	23	9500
10	33	15500
4	27	9300
12	31	26600
11,13	26	8450
15	25	23300
4	27	9300
17	29	6200
9,16	24	17700
15	25	23300
1	43	20900
8	28	8800
4	27	9300
8	28	8800
Average:	28.5	13982

2.2 Treatment effect estimates

- Differnece in average spending between treatment and control:
  - Before matching: $16426 - 20724 = -4298$
  - After matching: $16426 - 13982 = 2444$
- The example provides an ATET estimate: we find a match for each treated unit
- We can also use matching to estimate ATE and match in both directions:
  - If observation i is treated, we find matching partners from the untreated/control group
  - If observation i is untreated, we find matching partners from the treated group
  - Requires using observation multiple times
- Similarly, we can also focus on ATEU

2.3 Approximate matching

- Exact matches are often not available. In that case, a mathing estimator of ATET can also be constructed as:

$$\text{ATET} = \frac{1}{N_T} \sum_{W_i} (Y_i - Y_{j(i)})$$

  wjere $Y_{j(i)}$ is the outcome of an untreated observation such that $X_{j(i)}$ is the closest value to X_i among the untreated observations

- We may also use the average for M closest matches:

$$\text{ATET} = \frac{1}{N_T} \sum_{W_i} \left[Y_i - \left(\frac{1}{M} \sum_{m=1}^{M} Y_{j_m(i)} \right) \right]$$

- Works very well when we can find good matches for each treated unit, so that M is usually small because there are only few good matches

2.3.1 Bias of approximate matching

- Doing approximate matching will always be biased and may even be inconsistent
- The bias of the matching estimator depends on the matching discrepancies $||X_i - X_{j(i)}||$
- Having multiple matches may reduce variance but increases bias – bias-variance tradeoff
- There are bias correction techniques that can be applied but they typically have other limitations or strong assuptions

2.4 Important steps in implementing matching methods

Matching methods have five key steps:

- Design:
  1. Decide on which covariates X to include
  2. Defining "similarity" or "closeness"
     - Decide on a distance measure to determine similarity
  3. Implementing a matching method, given the similarity measure
     - How to assign control units to similar treatment units
  4. Assessing the quality of matching
     - Potentially iterating between steps 1 and 3 until well-matched sample results
- Analysis
  5. Estimate the treatment effect, given the matching done in step 3

2.4.1 Decide on which covariates to include

- Generally, we should include all pre-treatment covariates that potentially affect the treatment assignment and potential outcomes
  - Some methods pose limitations on the number of variables to include for matching
  - Inlcuding irrelevant variables may reduce quality of the matched samples
- We should not include covariates that are affected by the treatment

2.4.2 Decide on the distance measure

- There are different ways to define the distance D_{ij} between individuals i and j
- Exact matching:

$$D_{ij} = \begin{cases} 0 & \text{if } X_i = X_j \\ \infty & \text{if } X_i \neq X_j \end{cases}$$

  - May require binning or discretization of continuous covariates (e.g., age groups 20-30 etc)
  - For high dimensional X, exact matching often becomes infeasible (curse of dimensionality), and we need to rely on linear transformations of the X into a distance measure
- Mahalanobis:

$$D_{ij} = \sqrt{(X_i - X_j)'\Sigma^{-1}(X_i - X_j)}$$

  - More common than Euclidian or normed Euclidian distance for matching
  - Accounts for different variances of X and correlations between X
  - Works best with continuous covariates (assumes normally-, or t-distributed variables)
- Other common distance measures, like Manhattan, Chebychev, Euclidian

Some distance measures

Distance measure	Handles scale differences?	Accounts for correlation?	Use case / Advantage
Euclidean	No	No	When covariates have similar scale and are (relatively) uncorrelated
Normed Euclidean	Yes	No	When covariates have different scales (variances) but are (relatively) uncorrelated
Manhattan	No	No	When Euclidean distance over-penalizes large differences
Mahalanobis	Yes	Yes	When covariates are correlated and have different scales (variances)
Propensity score	Yes	No	When reducing high-dimensional covariates into a single balancing score; balance not required for all covariates but only for propensity score
...			

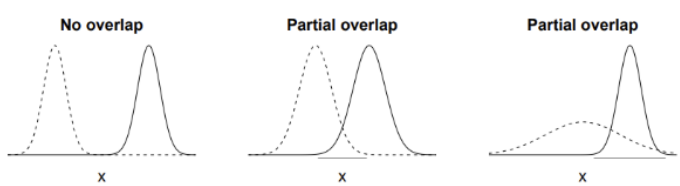
2.4.3 Implementing the matching method

- Greedy (nearest neighbor) or optimal matching?
- With or without replacement?
  - Reusing observations for more than one matching partner
  - Inference (standard errors) complicated to calculate
- Selecting the number of matches
  - Bias-variance tradeoff: Multiple controls increase bias since the 2nd, 3rd etc. matches are, by definition, further away from the treated individual than it is the 1st closest match. On the other hand, utilizing multiple matches can decrease variance due to the larger matched sample size
- Pruning
  - Defining a maximum distance for matching?
  - Avoids poor matches that can cause large bias
  - Loss of treatment units $\rightarrow$ not a true ATET anymore
- **Choices are interrelated:**
  - e.g., greedy is less of a problem when reusing observations (with replacement)
  - Purining is more important when not reusing observations

2.4.4 Assessing the quality of matching

Common support / overlap

- Common support is a prerequisite for matching: For every combination of covariates, there exists a positive probability of receiving each treatment level (i.e., there is no combination of covariates that prevents being in treated or untreated condition): $0 < \mathbb{P}[T_i = 1|X_i] < 1$
- We need to observe at least partial overlap of covariate distribution to be able to find matching parners (i.e., there are instances of X in both treatment and untreated condition)

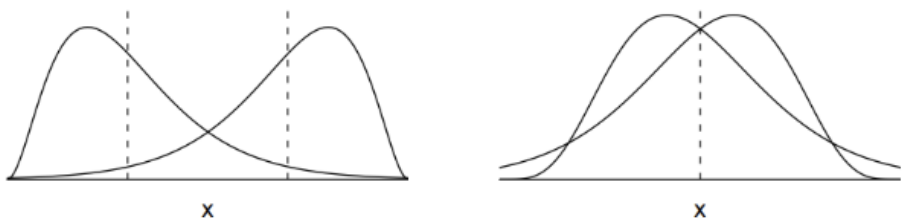


- Depends on whether we try to estimate ATE or ATET
  - For ATET we can discard control units but may not want to discard treatment units
  - For ATE we want to keep potentially all units and find a matching parner

2.4.5 Assessing the quality of the matched samples

Covariate balance

- After matching, the covariate distribution in the treated group should not differ from the untreated group
- Exact matching ensures covariate balance; all other methods need to be checked
- Not only equality of means, but also investigate the shape of the distribution (e.g., density histograms)



Both imbalance and lack of overlap are also problems for covariate adjustment in OLS regression, but balance may be harder to check than in matching

- OLS does not produce matched treated and untreated groups based on the covariates

2.5 Decide on the distance measure – alternatives

- Propensity score matching:

$$D_{ij} = |p(X_i) - p(X_j)|$$

  - The propensity score measures the probability of being treated conditional on X (next section)
- Logit propensity score matching:

$$D_{ij} = \Big|\text{logit}\big(p(X_i)\big) - \text{logit}\big(p(X_j)\big)\Big|$$

  - May provide better matching results than using propensity scores directly by ensuring that small differences in extreme propensity scores (close to 1 or 0) are deemphasized
- Combinations of exact matching and propensity score matching
  - Match on specific important covariates directly (e.g., loyalty card member)
  - Match within these strata using propensity score matching

3 Propensity scores

3.1 The core idea

- The propensity score measures the probability of being treated conditional on X

$$p(X_i) = \mathbb{P}[T_i = 1|X_i]$$

- In an ideal randomized experiment with two arms (i.e., $T_i \in \{0, 1\}$) and equal group sizes, the propensity score is constant at 0.5
  - More generally, for randomized T_i, we have $p(X_i) = p_0 \in \{0, 1\}$
- However,
  - We do not know the actual propensity scores $p(X_i)$ but can only estimate $\hat{p}(X_i)$
  - In practice, the $\hat{p}(X_i)$ is rarely constant even for randomized experiments because covariate distribution is rarely perfect in finite samples
  - Looking at the distribution/variability of the $\hat{p}(X_i)$, gives a measure of how far we are from a randomized experiment given the covariates
- Propensity scores are a great way to reduce the dimensionality of X because they have some nice properties
- Given the "selection on observables" (conditional independence) assumption:

$$\{Y_i(1), Y_i(0)\} \perp T_i | X_i$$

- Plus "common support"

$$0 < \mathbb{P}[T_i = 1|X_i] < 1$$

- Rosenbaum shows that this implies:

$$\{Y_i(1), Y_i(0)\} \perp T_i | p(X_i)$$

  - Conditioning on the propensity score is enough to have independence between the treatment indicator and the potential outcomes
  - If we are able to find exact matches in propensity scores, we have an unbiased estimator

3.2 Balancing properly

- T and X are independent conditional on $p(X)$: $X_i \perp T_i | p(X_i)$
  - Because $p(X)$ explains the selection mechanism into T given X
- Thus, we obtain the balancing propery of ther propensity score:

$$\mathbb{P}\big(X_i | T_i = 1,\, p(X_i)\big) = \mathbb{P}\big(X_i | T_i = 0,\, p(X_i)\big)$$

  - Conditional on the propensity score, the distribution of the covariates is the same for treated and untreated individuals
- We can use this to check if our estimated propensity score actually produces balance:

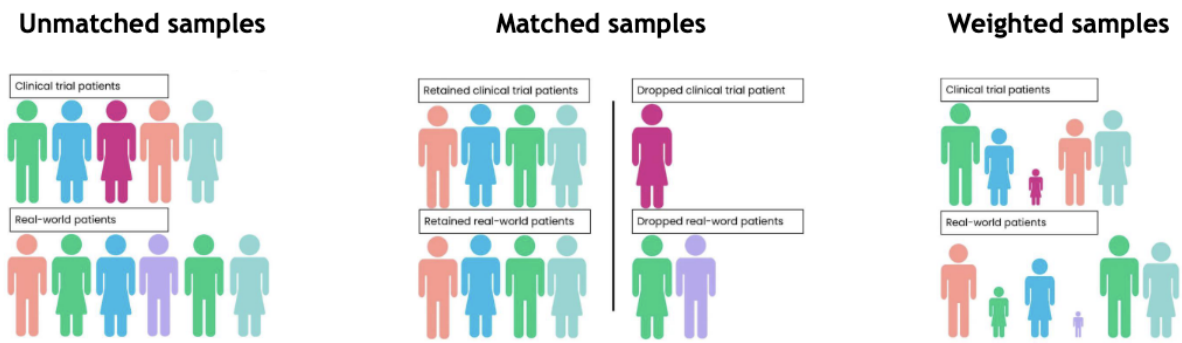
$$\mathbb{P}\big(X_i | T_i = 1,\, \hat{p}(X_i)\big) = \mathbb{P}\big(X_i | T_i = 0,\, \hat{p}(X_i)\big)$$

3.3 How to estimate the propensity score?

- Most commonly the propensity score $\hat{p}(X_i)$ is estimated through logistic regression
  - Limitations: specific functional form and parametric method
- Estimating the propensity score $\hat{p}(X_i)$ is a standard prediction problem for which we may use any nonparametric ML approach
  - e.g., classification trees, random forests, gradient boosting etc.
  - Requires measures to reduce overfitting
  - However, the predictive quality of the propensity score does not necessarily translate into its balancing properties; some research shows that fancy ML methods may not be better (but worse) than old-school logistic regression in reducing the confounding bias
- The task is not to best predict assignment, but to control for (include) only those variables that are predicting assignment and outcome (i.e., confounders)
  - Those that we need for conditional independence

4 Weighting

Weighting vs. matching



4.1 Inverse propensity weighting (IPW)

- Under the previous assumption $\{Y_i(1) - Y_i(0)\} \perp T_i | X_i$ implying $\{Y_i(1) - Y_i(0)\} \perp T_i | p(X_i)$
- The ATE can also be consistently estimated as

$$\text{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}\left[\frac{T_i}{p(X_i)} Y_i - \frac{(1-T_i)}{1-p(X_i)}\right] = \mathbb{E}\left[Y_i \frac{T_i - p(X_i)}{p(X_i)(1-p(X_i))}\right]$$

- Thus, given the estimated propensity scores $\hat{p}(X_i)$ for each individual, we can construct the so-called **inverse propensity weighted (IPW)** estimator $\hat{\tau}_{\text{IPW}}$

$$\hat{\tau}_{\text{IPW}} = \frac{1}{N} \sum_{i=1}^{N} \left[Y_i \left(\frac{T_i}{\hat{p}(X_i)} - \frac{(1-T_i)}{1-\hat{p}(X_i)}\right)\right] = \frac{1}{N} \sum_{i=1}^{N} \left[Y_i \frac{T_i - \hat{p}(X_i)}{\hat{p}(X_i)(1-\hat{p}(X_i))}\right]$$

  - Weights up (or down) the over-represented (or under-represented) observations

4.2 Propensity score weighting – numerical examples

W	$p(X_i) = \mathbb{P}[W_i = 1 \mid X_i]$	ipw
1	0.99	1.010
1	0.75	1.333
1	0.50	2
1	0.25	4
1	0.01	100
1	...	...
0	0.99	-100
0	0.75	-4
0	0.50	-2
0	0.25	-1.333
0	0.01	-1.010
0	...	...

- Large propensity score and treated ($T = 1$)
  - The likelihood is high that those observations got treatment based on the X and thus do not need (much) reweighting
  - Low weight (close to one)
- Low propensity score and treated ($T = 1$)
  - The likelihood is low that those observations got treatment based on the X (we have only few examples in the data) and thus these need reweighting (need to get a larger weight to make the overall sample balanced)
  - Large weight
- Vice versa for untreated

4.3 Issues with propensity score weighting

- Core issue with matching estimators remains: require balance and overlap
- IPW has an inherent balancing property, but still requires overlap in the propensity score distribution
- When overlap is weak, models rely on extrapolation in regions where data are sparse
- Large variance: the estimator typically has quite large variance and is not very stable

4.4 Doubly robust estimators (AIPW)

- Combines propensity score based and regression-based estimation
- Estimators that only need the propensity score OR the regression function to be correctly specified: the augmented IPW estimator (AIPW)

$$\text{ATE} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{T_i}{\hat{p}(X_i)}\left(Y_i - \hat{\mu}_{T=1}(X_i)\right) + \hat{\mu}_{T=1}(X_i)\right) - \frac{1}{N}\sum_{i=1}^{N}\left(\frac{(1-T_i)}{1-\hat{p}(X_i)}\left(Y_i - \hat{\mu}_{T=0}(X_i)\right) + \hat{\mu}_{T=0}(X_i)\right)$$

  - With $\hat{\mu}_{T=1}(X_i)$ and $\hat{\mu}_{T=0}(X_i)$ being the predicted value from a regression of X on Y for only treated or untreated observations
- Typical way to estimate standard errors / confidence intervals: bootstrap
- To better understand: rewrite in two parts:

$$\text{ATE} = \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{\mu}_{T=1}(X_i) - \hat{\mu}_{T=0}(X_i)\right)}_{\text{Simple regression adjustment}} + \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{T_i}{\hat{p}(X_i)}\left(Y_i - \hat{\mu}_{T=1}(X_i)\right) - \frac{(1-T_i)}{1-\hat{p}(X_i)}\left(Y_i - \hat{\mu}_{T=0}(X_i)\right)\right)}_{\text{IPW estimator on the residuals of the regression model } Y_i - \hat{\mu}(X_i)}$$

- If the regression model is correct, then $\mathbb{E}[Y_i - \hat{\mu}(X_i)|X_i]$ will be zero and the second part vanishes (i.e., we don't need the second part, because the first is already giving the right estimate)
- If the regression model is not correct, but the propensity score model is, the second part will properly correct (debias) the regression model estimates using the residuals
- However, $\hat{\mu}_{T=1}(X_i)$ and $\hat{\mu}_{T=0}(X_i)$ must not come from an OLS regression

4.5 Characteristics of the AIPW model

- Model misspecification
  - The estimator is doubly robust: it requires only one of the two models – propensity score model or outcome model to be correct
  - However, if both are wrong, the bias can be larger than in any of the two separately
- Variance and instability in small samples
  - Usually, AIPW has smaller variance than the IPW estimator
  - If the outcome model is overfitted, it can lead to poor generalization